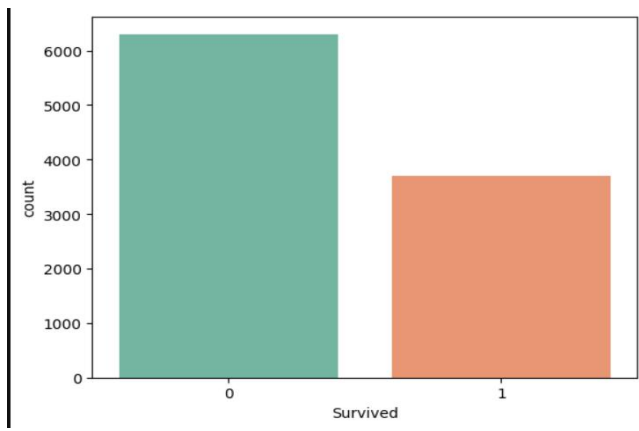
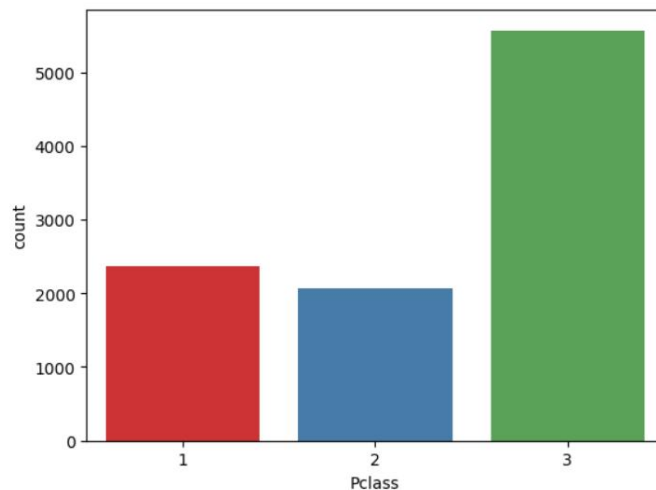


Task 5: Exploratory Data Analysis (EDA):

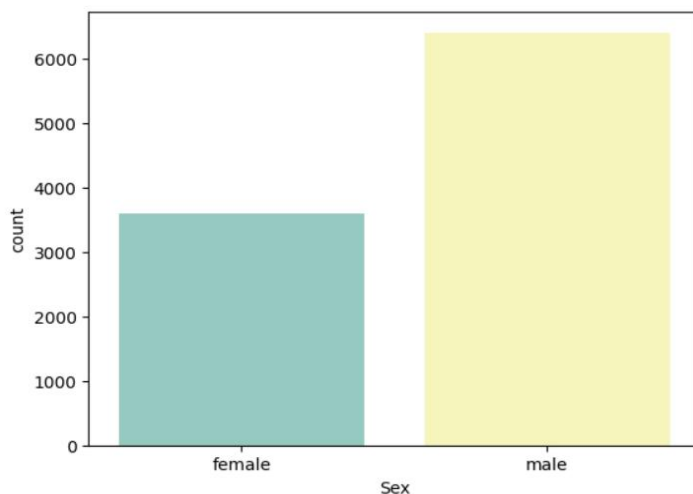
Visual Observations:



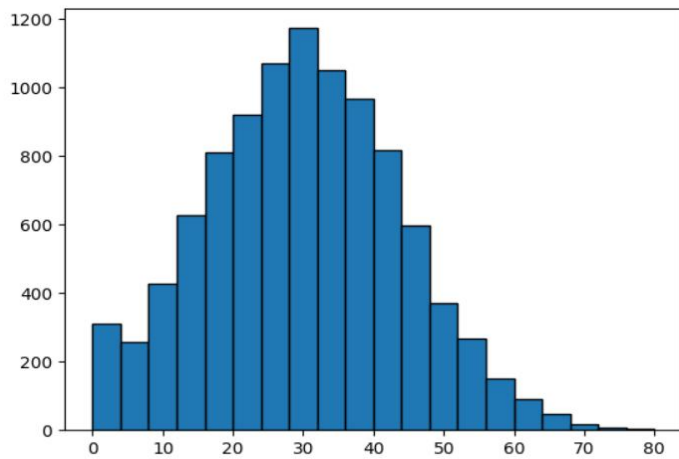
More passengers did not survive (Survived = 0) than those who did. This class imbalance is important for Modeling.



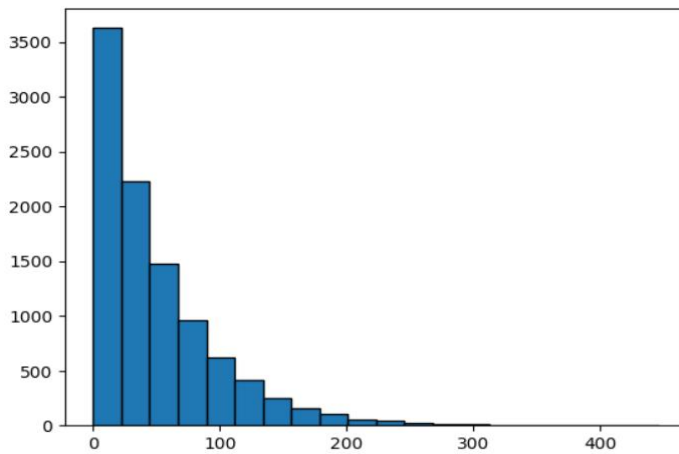
Most passengers were in 3rd class, fewer in 1st class. This may reflect socio-economic demographics on the Titanic.



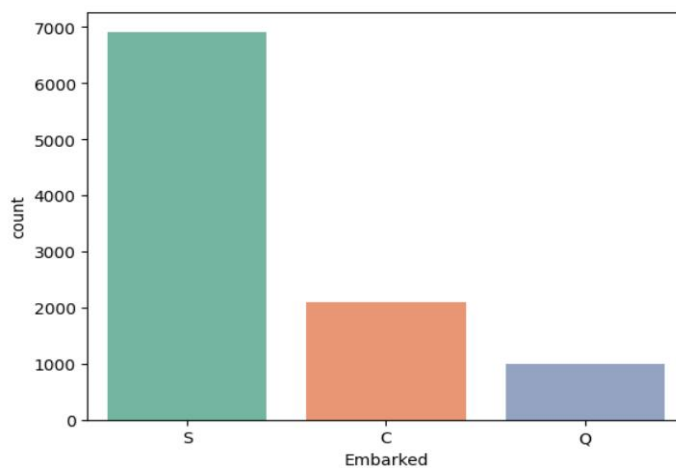
Females had a much higher survival rate than males. Most males did not survive. This visual is about Not survived People. Here males did not Survived More.



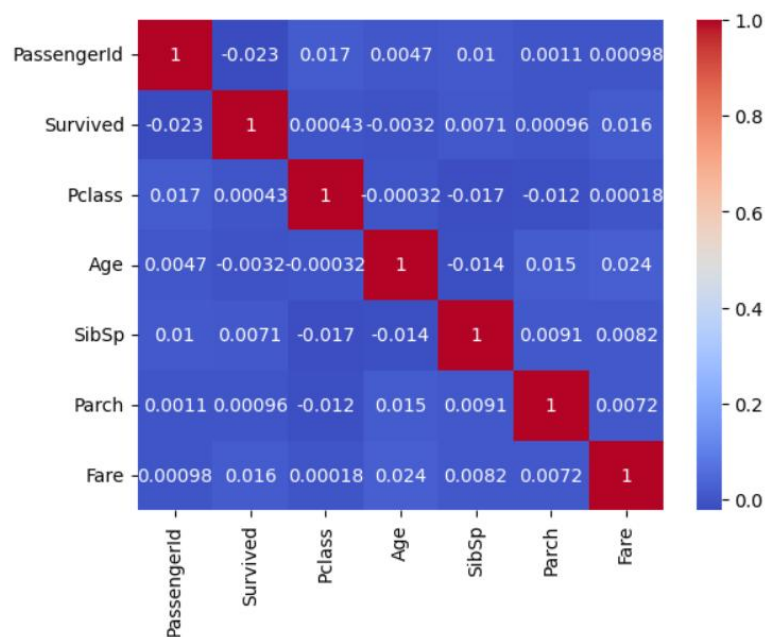
Most passengers were between 20–40 years old.
 The distribution is right-skewed (a few very old passengers).
 There were some infants and children as well.



Most fares were below \$100.
 Distribution is highly right-skewed, with a few passengers paying extremely high fares.



Most passengers boarded from Southampton ("S").
 Survival was highest from Cherbourg ("C"), possibly due to more first-class passengers.



Fare and Pclass are negatively correlated (richer people tend to be in higher classes).

Survived shows:

Positive correlation with Fare and Pclass (1st class more likely to survive).

Negative correlation with Sex if encoded (male = 1, female = 0).

Summary of Findings:

Feature	Finding
Sex	Females had a significantly higher survival rate than males.
Pclass	1st class passengers were more likely to survive. 3rd class had the lowest survival.
Fare	Higher-paying passengers had better chances of survival.
Embarked	Cherbourg had higher survival rates.
Age	Children and young adults had a mix of survival outcomes; infants had slightly higher survival.
Cabin	Often missing; useful for extracting deck info. Can correlate with survival.
Correlations	Survival is positively linked with Fare, negatively with being male, and positively with being in a higher class.